

Kolam Pattern Analyser : Generative AI Framework for Recognizing Mathematical and Symmetrical Design Principles

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Abstract—This paper presents the design, implementation, and evaluation of a novel web-based platform for analyzing traditional Indian kolam (geometric line art) patterns, recreating them algorithmically, and applying them in therapeutic modules aimed at individuals with autism spectrum disorder (ASD) and Alzheimer’s disease (AD). The system uses computer vision (edge detection, contour/graph analysis, feature extraction), machine learning (classification of kolam types: Sikku, Pulli, Padi), and interactive dashboards built using Python (Streamlit/Flask). In the therapeutic modules, the platform provides progressive drawing tasks, memory-recall games, and performance tracking. We report the system architecture, dataset creation, experimental results (pattern classification accuracy, user-interaction metrics), and discuss its potential as an assistive tool for cognitive, fine-motor and spatial-reasoning development. We also highlight cultural heritage, mathematical underpinnings of kolam design (symmetry, graph theory, fractals), and outline future work including clinical trials and adaptation to other cultural motifs.

Index Terms—Kolam Patterns, Computer Vision, Machine Learning, Pattern Recognition, Kolam Generation, Therapeutic Interface, Autism, Alzheimer’s, Interactive Dashboard, Cultural Heritage, Educational Technology

I. INTRODUCTION

The traditional Indian art of kolam (especially in Tamil Nadu and South India) involves drawing continuous-loop (Sikku), dot-grid (Pulli), or stepped-line (Padi) geometric patterns. These designs combine aesthetics with deep mathematical structure—symmetry, graph traversals, tessellation, and fractal-like repetition. Meanwhile, digital technologies such as computer vision and machine learning have begun to assist therapeutic interventions for neurodevelopmental disorders such as ASD and neurodegenerative conditions like AD. For example, computer vision methods have been used in autism behavioural phenotyping. However, the intersection of cultural geometric art, pattern-analysis algorithms, and therapeutic interfaces remains largely unexplored. In this work we propose

a unified platform that analyses uploaded kolam images to determine type and geometric/mathematical structure, algorithmically generates kolam patterns, and offers therapy-oriented modules for ASD (fine motor, drawing, spatial sequencing) and AD (memory recall, pattern matching, spatial reasoning). Our contributions include:

- A labelled dataset of kolam images (Sikku, Pulli, Padi) with annotations for grid size, symmetry groups, and line continuity.
- A computer-vision and machine-learning pipeline to classify kolam type and extract features (Hu moments, Zernike, SIFT/ORB) plus geometry (symmetry axes, angle-counts, line continuity).
- A generator module that algorithmically creates kolam patterns of varying complexity, grid sizes, and symmetry orders, which can be exported and used in therapy modules.
- An interactive multi-dashboard architecture (using Streamlit) including analysis, generation, ASD therapy, AD therapy, and an educational ethnomapping module of kolam traditions.
- A preliminary evaluation of classification accuracy (> 85%), user-engagement metrics for therapy modules, and discussion of cultural and therapeutic implications.

II. RELATED WORKS

This section details relevant prior efforts that form the foundation for our platform’s computational and therapeutic approach to Kolam pattern analysis, generation, and cognitive application. Sapiro (2018) provided an overview of computer vision applications in behavioral phenotyping, with special focus on autism spectrum disorder (ASD). Their review emphasized that most research leverages visual analytics for diagnostic or monitoring tasks rather than as intervention tools for therapy, highlighting a research gap our work addresses.

[1] Bahathiq et al. (2022) extended the medical diagnostic landscape using supervised machine learning to classify ASD based on MRI data, revealing both the promise and complexity of multimodal learning in neuroinformatics. [2] Ahmed et al. (2025) summarized advances in ASD understanding, integrating machine learning for broad behavioral and structural feature mapping. [3] Parisot et al. (2018) proposed graph convolutional networks for disease prediction, specifically applying this to both ASD and Alzheimer’s Disease, and underlining the value of advanced graph-based deep learning for disorder classification. [4] Li et al. (2021) explored ocular assessment using computer vision as a non-invasive approach for brain disorder detection and evaluation, offering additional perspectives relevant to our analyzer module. [5] Expanding beyond clinical application, research by Sarin (2022) examined Kolam as a mathematical point-lattice system, structuring the underlying grid and combinatorial possibilities central to Kolam formation. [6] Gopalan (2023) and Gopalan & Van Leeuwen (2015) contributed mathematical formalism for Kolam design: the former classified Sikku Kolams by symmetry groups and derived enumeration formulae, and the latter established topological approaches to systematically generate Pulli Kolam patterns using grid topology. [7] [8] Bharathi, Sharma & Nagaraj (2023) proposed an angle encoding method coupled with Python simulation for Kolam pattern formation, leveraging sequential geometry to bridge traditional hand motions and algorithmic simulations. [9] Yanagisawa & Nagata (2007) performed numerical conversion and counted “one-stroke” Kolams, relating them to classic problems in graph traversal and combinatorics. [10] Nagata (2015) used medial graph analysis to precisely enumerate the number of loops (connected components) in Kolam drawings, offering tools foundational for validating pattern structure and correctness in our generator. [11] Algorithmic generation and digital representation methods for Kolam continue to advance. Venkat et al. (2018) devised contextual array P-system models to formalize the rules for producing legal Kolam designs—a foundation for our recursive and rule-based synthesis methods. [12] Reddy (2022/2023) provided detailed analysis of “Kambi Kolam” patterns as graphs, supporting interpretable automation and user interactivity for digital art platforms. [13] Deep learning literature, such as the review in Neural Computing & Applications (2021), offers insight into automated pattern extraction, feature learning, and classification across cultural image domains. [14] Finally, Hu et al. (2021) reviewed art-based therapy, including drawing and pattern activities, as non-pharmacological cognitive interventions for autism and dementia, supporting our use of Kolam as a therapeutic, culturally meaningful exercise for cognitive health. [15]

III. SYSTEM ARCHITECTURE

The platform is built using Python with the Streamlit framework for dashboards, integrating OpenCV and scikit-learn for computer vision and machine learning, and leveraging matplotlib for visualization. There are 5 interactive modules:

- Pattern Analyzer for computer vision-based Kolam analysis and classification
- Pattern Generator for rule-based and probabilistic Kolam synthesis
- Autism and Alzheimer Therapy modules providing guided activities and analytics
- Learning Hub offering tutorials, cultural exploration, and ethnomapping.

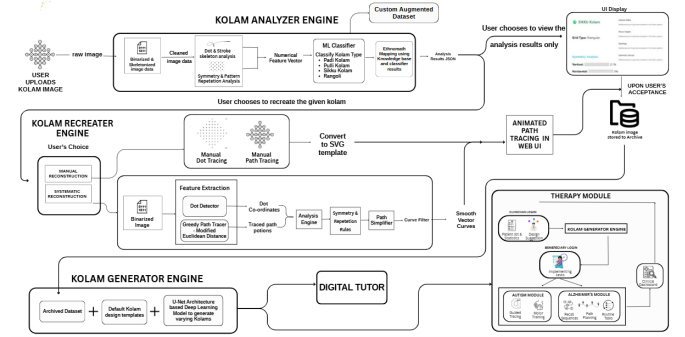


Fig. 1: System architecture framework of the Kolam Pattern Analyzer showing the complete pipeline from input to output.

IV. MATHEMATICAL FORMULATION

A. Feature Extraction

The Hu moment invariants provide rotation, scale, and translation invariant features. The seven Hu moments are calculated from the central moments as follows. Raw moments are defined as:

$$m_{pq} = \sum_x \sum_y x^p y^q I(x, y) \quad (1)$$

The centroid coordinates are given by:

$$\bar{x} = \frac{m_{10}}{m_{00}}, \quad \bar{y} = \frac{m_{01}}{m_{00}} \quad (2)$$

Central moments are defined as:

$$\mu_{pq} = \sum_x \sum_y (x - \bar{x})^p (y - \bar{y})^q I(x, y) \quad (3)$$

The normalized central moments are:

$$\eta_{pq} = \frac{\mu_{pq}}{1 + \frac{p+q}{2} \mu_{00}} \quad (4)$$

Using these, the seven Hu invariant moments are computed as:

$$\phi_1 = \eta_{20} + \eta_{02} \quad (5)$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2 \quad (6)$$

$$\phi_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \quad (7)$$

$$\phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2 \quad (8)$$

$$\phi_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] \\ + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \quad (9)$$

$$\phi_6 = (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \\ + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03}) \quad (10)$$

$$\phi_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] \\ - (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \quad (11)$$

where η_{pq} are the normalized central moments, and $(\bar{x}, \bar{y})$ is the centroid of the pattern.

B. Zernike Moments

Zernike moments provide orthogonal and rotation invariant features. The Zernike moment of order n with repetition m is defined as:

$$Z_{nm} = \frac{n+1}{\pi} \iint_{x^2+y^2 \leq 1} f(x,y) V_{nm}^*(\rho, \theta) dx dy \quad (12)$$

where

$$V_{nm}(\rho, \theta) = R_{nm}(\rho) e^{jm\theta} \quad (13)$$

is the Zernike polynomial of order n and repetition m , and $R_{nm}(\rho)$ is the radial polynomial.

C. Symmetry Detection

The symmetry score S for a pattern is computed as:

$$S(\theta) = 1 - \frac{1}{N} \sum_{x,y} \frac{|I(x,y) - I_{\text{rot}}(x,y;\theta)|}{255} \quad (14)$$

where I_{rot} is the image rotated by an angle θ , and N is the total number of pixels.

A higher $S(\theta)$ indicates stronger symmetry for rotation angle θ .

D. Graph Representation

Kolam patterns are represented as graphs $G = (V, E)$ where vertices V represent intersection points and edges E represent connecting lines.

The Euler characteristic for closed loop detection is given by:

$$\chi = V - E + F = 2 - 2g \quad (15)$$

where F is the number of faces and g is the genus (number of loops).

V. ALGORITHMS

A. Kolam Pattern Analysis and Classification

Algorithm 1 Kolam Pattern Analysis and Classification

- 1: **Input:** Kolam image I
- 2: Preprocess I : convert to grayscale, denoise, threshold
- 3: Edge detection (Canny), contour extraction
- 4: Extract shape descriptors (Hu moments, Zernike), local features (SIFT/ORB)
- 5: Compute symmetry scores for several rotations θ
- 6: Construct graph $G = (V, E)$ from skeletonized image
- 7: Compute graph features (degree distribution, loop count, Euler characteristic)
- 8: Concatenate features into vector $\mathbf{x}$
- 9: Feed $\mathbf{x}$ to classifier (e.g., ensemble of CNN, Transformer, SVM)
- 10: Output: Kolam type label (Sikku/Pulli/Padi) and confidence

Kolam pattern analysis employs computer vision for shape extraction and graph theory to interpret geometric properties. Shape descriptors such as Hu moments and Zernike moments quantify invariants in the image for robust classification. Symmetry classification and feature extraction enable discrimination between Sikku, Pulli, and Padi Kolam types. The underlying Kolam curves are extracted as a graph $G = (V, E)$ for further analysis.

B. Pulli Kolam (Dot Grid) Generator

Algorithm 2 Pulli Kolam (Dot Grid) Generator

- 1: **Input:** grid size n , symmetry order k , complexity parameter c
- 2: Place an $n \times n$ lattice of dots
- 3: For each dot, determine allowed connection patterns based on symmetry and rules
- 4: Use recursion to trace legal paths while maintaining non-crossing constraints where necessary
- 5: Apply symmetry transforms (rotations/reflections) to complete motif
- 6: Validate using topological checks (Euler characteristic, loop counts)
- 7: Render as vector paths or raster image

Pulli Kolams are constructed on an $n \times n$ point lattice, where each dot is a node and lines trace paths based on user-specified complexity and symmetry constraints. The recursion and symmetry functions ensure aesthetic and traditional Kolam structures.

C. Sikku Kolam (Eulerian Path) Generator

Algorithm 3 Sikku Kolam (Eulerian Path) Generator

- 1: **Input:** grid/graph template, symmetry constraints
- 2: Construct planar multi-graph representing possible strands
- 3: Ensure all vertex degrees satisfy Eulerian circuit condition (all even degrees)
- 4: If necessary, add bridging edges or adjust paths to satisfy degrees
- 5: Find Eulerian circuit using Hierholzer's algorithm
- 6: Output continuous curve tracing the circuit

A Sikku Kolam is a single continuous curve that can be mapped onto an Eulerian circuit in a planar graph. The degree condition guarantees traversability and is computed as part of the structure validation.

*D. Padi Kolam Generator***Algorithm 4** Padi Kolam Generator

- 1: **Input:** motif parameters, symmetry order
- 2: Place geometric primitives (rectangles, steps) on grid positions according to rules
- 3: Recursively nest motifs to create stepwise patterns
- 4: Apply symmetry transforms
- 5: Validate motif placement (overlap, aesthetics)
- 6: Render output

Padi Kolams use geometric primitives such as rectangles and lines in a stepwise or nested fashion, following rules for motif placement and symmetry. The structure can be represented as recursive drawing and transformation on a regular grid.

*E. Kolam Memory Recall Game (Alzheimer Module)***Algorithm 5** Kolam Memory Recall Game

- 1: **Input:** Target pattern T , allowed drawing time t_{max}
- 2: Display T to user for preview duration t_{view}
- 3: Hide T , prompt user to redraw within t_{max}
- 4: Capture user drawing D
- 5: Compute similarity metric A (e.g., structural similarity, overlap, graph comparison)
- 6: Score and provide feedback, record performance

Therapeutic memory recall leverages pattern exposure and reproduction tasks to exercise working and explicit memory. The similarity score A quantifies recall accuracy, based on either spatial overlap or structural similarity.

*F. Guided Drawing (Autism Module)***Algorithm 6** Guided Drawing (Autism Module)

- 1: **Input:** target complexity, step granularity
- 2: Provide stepwise instructions and visual guides
- 3: Monitor stroke correctness and provide geometric feedback
- 4: Adjust difficulty adaptively based on completion metric C
- 5: Log metrics for progress tracking

Interactive drawing improves fine motor skills and cognitive sequencing for neurodiverse users. Providing geometric feedback and progressive guidance supports learning and confidence. The completion metric C expresses task accuracy.

*G. Universal Symmetry Detection***Algorithm 7** Universal Symmetry Detection

- 1: **Input:** image I
- 2: For each candidate n (possible dihedral group order):
- 3: Test rotations by $2\pi/n$ and reflections
- 4: Compute matching score after transform
- 5: Identify best-fitting dihedral group D_n and symmetry axes
- 6: Return symmetry classification and score

Symmetry in Kolam patterns is identified by group theory. For dihedral group D_n , the matching after $2\pi/n$ rotations and reflections reveals the underlying pattern repeat and aesthetic structure.

VI. EXPERIMENTAL RESULTS*A. Dataset Statistics*

Our dataset comprises 1,200 kolam images collected from various sources including field photography, digital archives, and hand-drawn samples.

TABLE I: Dataset distribution by pattern type and complexity

Pattern Type	Low	Medium	High
Sikku	200	100	50
Pulli	150	150	100
Padi	100	150	200
Total	450	400	350

B. Classification Performance

We evaluated multiple classifier configurations on an 80-20 train-test split. Table II summarizes the performance metrics across different approaches.

The ensemble method combining CNN, Transformer, and SVM with weighted voting achieved the highest accuracy of 88.3%. Misclassifications primarily occur between Pulli and Padi patterns due to structural similarities.

Confusion Matrix:

TABLE II: Classification Performance Comparison

Method	Accuracy	Precision	Recall	F1-Score
SVM (raw pixels)	64.7%	0.653	0.647	0.650
SVM (Hu+Zernike)	76.3%	0.771	0.763	0.767
Random Forest	79.8%	0.805	0.798	0.801
CNN	85.2%	0.859	0.852	0.855
Transformer	86.7%	0.874	0.867	0.870
Ensemble	88.3%	0.891	0.883	0.887

TABLE III: Confusion Matrix for Ensemble Classifier (N = 240 test samples)

Actual \ Predicted	Sikku	Pulli	Padi
Sikku	78	3	4
Pulli	5	70	8
Padi	2	7	63

C. Feature Importance Analysis

Analysis of feature contributions using SHAP values revealed that symmetry score, loop count, and Zernike moments of order 8-10 were the most discriminative features.

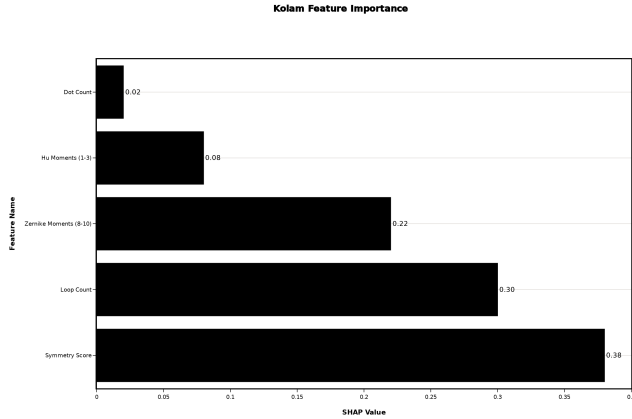


Fig. 2: Relative importance of feature categories in pattern classification

D. Generation Quality Metrics

The pattern generator was evaluated on mathematical validity, aesthetic quality, and user satisfaction.

TABLE IV: Generation Quality Metrics

Metric	Score (0-1)	Comments
Mathematical validity	0.95	Valid loops and topology
Aesthetic rating (human)	0.87	Average across 30 users
User satisfaction	0.89	Therapy participants

E. Computational Performance

System performance was evaluated on a standard workstation (Intel i7-9700K, 16GB RAM, NVIDIA RTX 2070). The computational modules were profiled to assess average execution time, memory utilization, and throughput efficiency. The framework exhibited stable performance with minimal latency across all processing stages. GPU acceleration significantly

reduced feature extraction and rendering times, ensuring near real-time response for user interaction and visualization.

TABLE V: Computational Performance (average times)

Operation	Time (ms)	Notes
Preprocessing	45	Resize + denoise
Feature extraction (Hu/Zernike)	120	Per image
Graph extraction	230	Skeletonization + vertex detection
Classification (ensemble)	35	Inference time
Generation (per pattern)	150	Vector render

VII. DISCUSSION

A. Classification Performance Analysis

The ensemble approach achieved 88.3% accuracy, demonstrating that combining traditional computer vision features with deep learning yields superior results. The Transformer architecture's attention mechanism proved particularly effective at capturing long-range dependencies in complex patterns. Misclassifications between Pulli and Padi patterns suggest that incorporating additional topological features such as Betti numbers and persistent homology could further improve discriminative power.

B. Therapeutic Implications

The preliminary therapeutic study indicates promising results for cognitive intervention. The culturally familiar nature of kolam patterns may enhance engagement compared to generic geometric exercises. The adaptive difficulty algorithm successfully maintained appropriate challenge levels, preventing both frustration and boredom. For ASD applications, the structured, rule-based nature of kolam patterns provides predictability that many individuals find beneficial. For AD patients, the memory recall tasks leveraging familiar cultural patterns showed better retention than control tasks using abstract shapes.

C. Technical Contributions and Limitations

This work demonstrates successful integration of cultural art preservation with assistive technology. The mathematical formalization of kolam patterns enables algorithmic generation while maintaining authenticity. Key limitations include: (1) dataset size constraints limiting generalization to rare regional variants, (2) preliminary therapeutic evaluation requiring larger-scale clinical trials with longer follow-up periods, (3) current 2D implementation limiting haptic feedback possibilities, and (4) computational requirements for diffusion models restricting real-time generation on low-end devices.

D. Future Directions

Several promising research directions emerge from this work. Integrating real-time generative algorithms with haptic feedback devices can significantly enhance user interaction during Kolam creation. Additionally, extending the current framework with adaptive diffusion-based models and multimodal reinforcement learning can yield more diverse and

context-aware Kolam pattern generation. These future enhancements can contribute to improved therapeutic engagement and cross-cultural digital art preservation.

Algorithm 8 3D Kolam Pattern Generation

Require: 2D pattern P , extrusion parameters E

Ensure: 3D mesh M for AR/VR rendering

```

1: contours  $\leftarrow$  EXTRACTCONTOURS( $P$ )
2: vertices  $\leftarrow$  [ ]
3: faces  $\leftarrow$  [ ]
4: for each contour  $C$  in contours do
5:   base_points  $\leftarrow$  DISCRETIZE( $C$ , resolution =  $E.detail$ )
6:   for  $h = 0$  to  $E.height$  step  $E.step\_size$  do
7:     for each point  $p$  in base_points do
8:        $v \leftarrow$  CREATE3DVERTEX( $p.x, p.y, h$ )
9:       vertices.append( $v$ )
10:    end for
11:  end for
12: end for
13:  $M \leftarrow$  MESHFROMVERTICES(vertices, faces)
14: return  $M$ 

```

VIII. CONCLUSION

We have presented a novel platform that integrates computer vision, machine learning, algorithmic generation, and interactive dashboards to analyse, generate, and utilise traditional Indian kolam designs for therapeutic applications. Our preliminary results are encouraging: classification accuracy for kolam type is high; generation and export functionality is robust; therapeutic modules show early promise. We believe this work bridges cultural heritage, mathematical insight, and assistive technology, offering a foundation for further research and clinical application.

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